Aggr Assignment-02

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Section: 05

faculty: AAD

CSE:-320

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1. Data communication is a exchange of data between two devices Sender and receiver through a medium/channel following certain rules/protocols.

There are four fundamental characteristics:

a) Delivery: The system must deliver data to correct destination. Data must be received by the intended devices or user and only by that device or user.

b) Accuracy: The system must be deliver the data accurately. Data that have been altered in ~~test~~ transmission and test uncorrected are unstable.

c) Timeliness: The system must deliver in a timely manner. Data delivered late are useless.

d) Jitter: Jitter refers to the variation in the packet arrival time. It is the uneven delay in the delivery of ~~vd~~ audio or video packets.

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2. we represent data such as text, numbers, image as audio and video.

Text: sequences of characters, including letters, numbers symbols. and spaces are represented in the text data.

Audio: Sound wave are represented as audio data, which can be stored in WAV, MP3 and AAC formats.

Videos: Sequences of images (frames) captured over the time are reprentend as video data, which can be recorded in AVI, MPEG and MP4 foromates.

Numeric: Numbers are used to repoesent numeris data. which can be a variety of forms including interger, floating-plot and fixed-point.

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2. Encryption: Encryption occours at the presentation Layer.

Error control: It is in the Transport Layer. Protocols like TCP provide error control.

connection control: connection control managed at the Transport Layer. Protocol like TCP establish and manage connections between devices.

Sessions management: Session management is a function of the Sesson Layer of the OSI model. The Layer establish, maintainee of sessions between applications.

Logleatl Location mangangement: Network layers is the respons foro the logleal location mangangement.

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Mac Address (Media Access control): Mac addresses are hardware addresses that are unique to each network interface card. Mac addresses of ~~the~~ a device never change.

IP address (Internet Protocol): IP address are logical addresses assigned to devices on a network. They can not change as data packets travel across different networks. IP addresses of a device is change by network.

MAC Address and IP Address are never change.

Answer to the Question-03

Baseline wandering is a average signal power at the receiver is taken as baseline, value of data element at the reciever is determind against this baseline reference. If there are ~~a~~ long of ones and zeros in the data, it drifts baseline and makes it tough for the receiver to decode the data correctly. This drift in the baseline is known as "baseline wandering".

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c)

DC component : Every Symbol used in the communication should have a signal average of 0 for the duration of the symbol. ~~A signal~~

Scrambling is a technique that does not increase the number of bits and does provide synchronization.

Purpose: To reduce the length of strings 0s or 1s in a transmitted signal.

Answer to Question-09

Scrambling is the transformation of data when their is a long sequence of zero's or one's. ~~This~~

→ 1010000000000

B8ZS replaces 8 consective 0s in sequence with

000VB0VB

a) 1010000000000

B8ZS : 101000VB0VB

HDB3 : 101B00V ⇒ even.

b) B8ZS: 101 000BV0VB

HDB3: 101B00V

Answer to the question-08

The Advantages of Block coding:

1. Block coding ensure Syeronization.
2. Improve the performance of line coding
3. Direct error in the transmission.

The disAdvantages of Block coding:

1. Does not Solve the DC component of NRZ-I.
2. latency in communication system.

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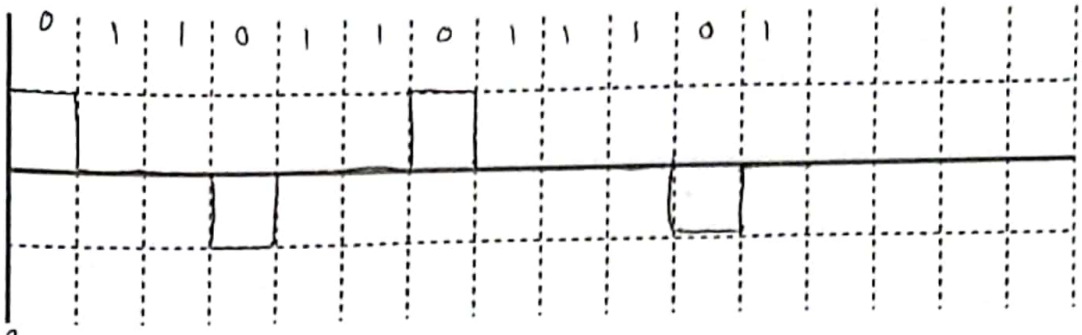
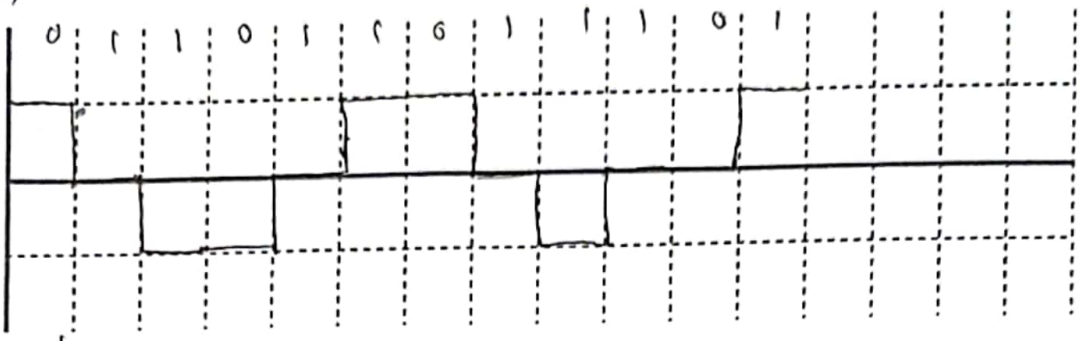
Pseudoternary

a)

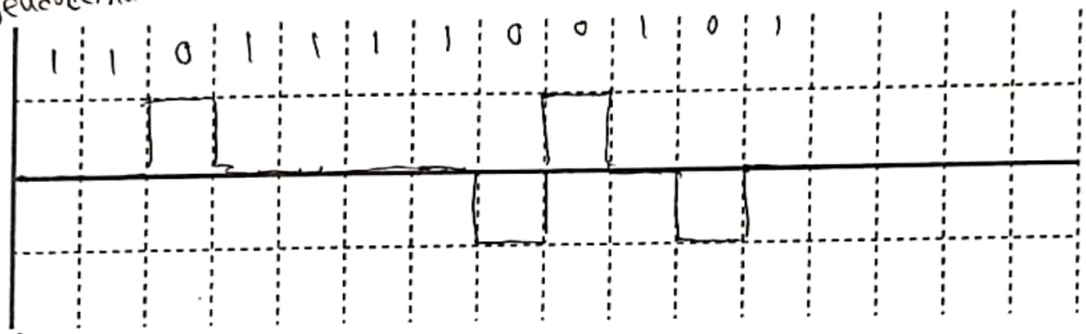


MLT-3

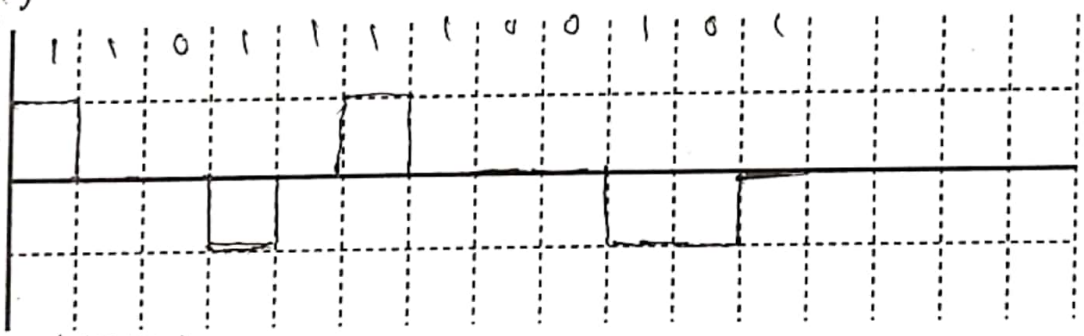


pseudoternary

b)

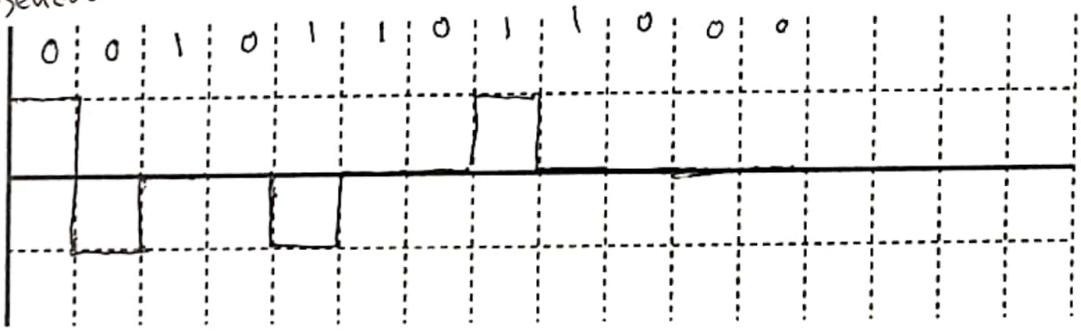


MLT-3



pseudoternary

c)



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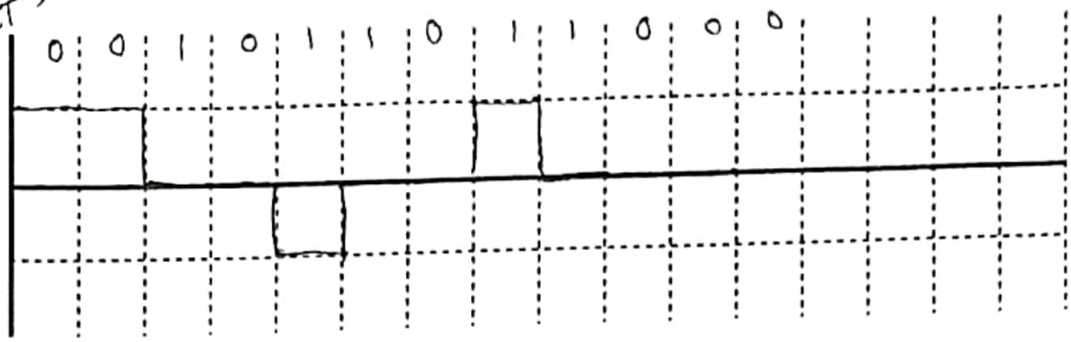
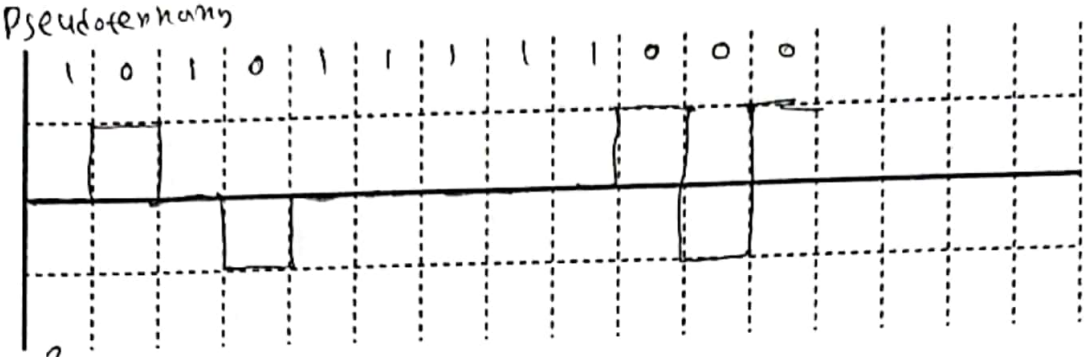
MLT-3

0 0 1 0 1 1 0 1 1 0 0 0



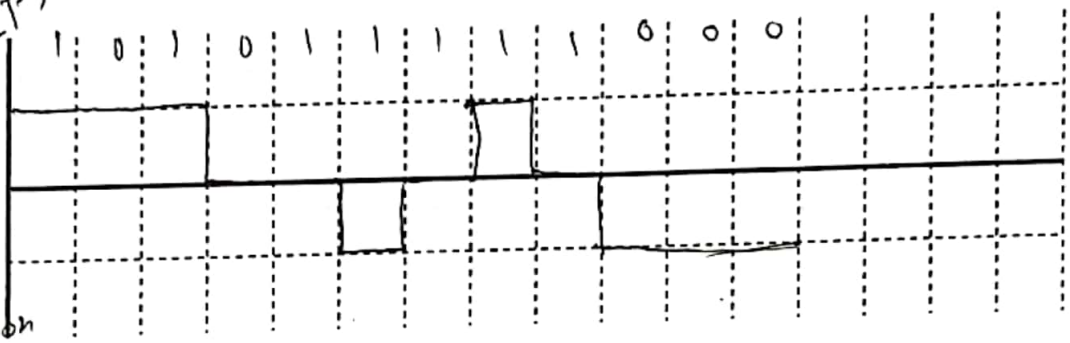
d) Pseudoternary

1 0 1 0 1 1 1 1 1 0 0 0



MLT-3

1 0 1 0 1 1 1 1 1 0 0 0

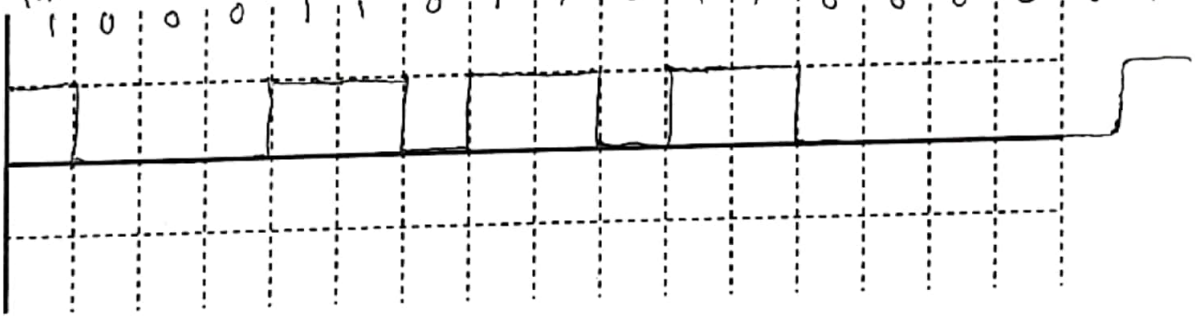


ZNO question

NRZ

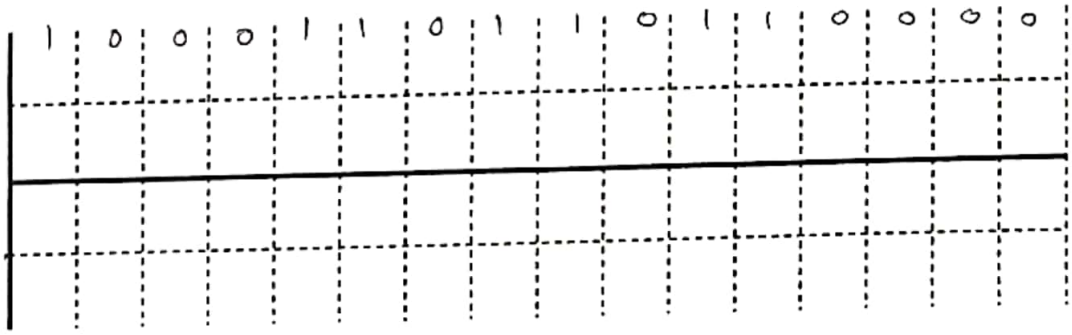
1 0 0 0 1 1 0 1 1 0 1 1 0 0 0 0 0 1

a)



1 0 0 0 1 1 0 1 1 0 1 1 0 0 0 0 0 1

b)



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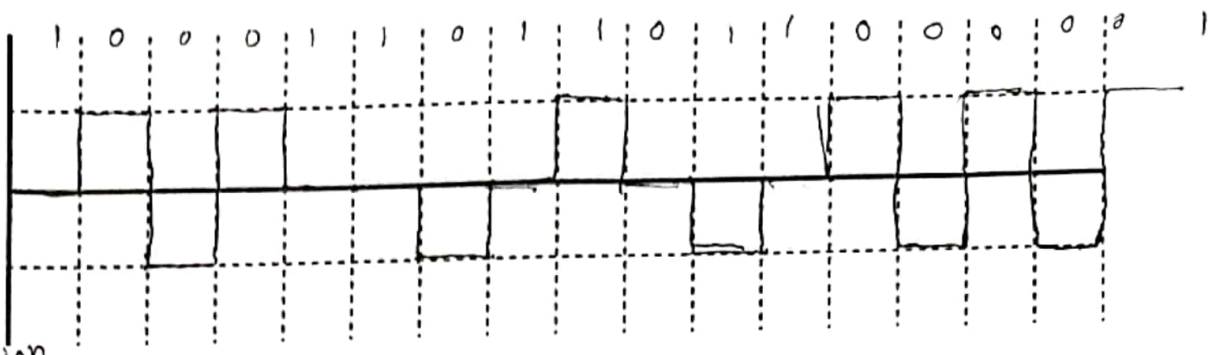
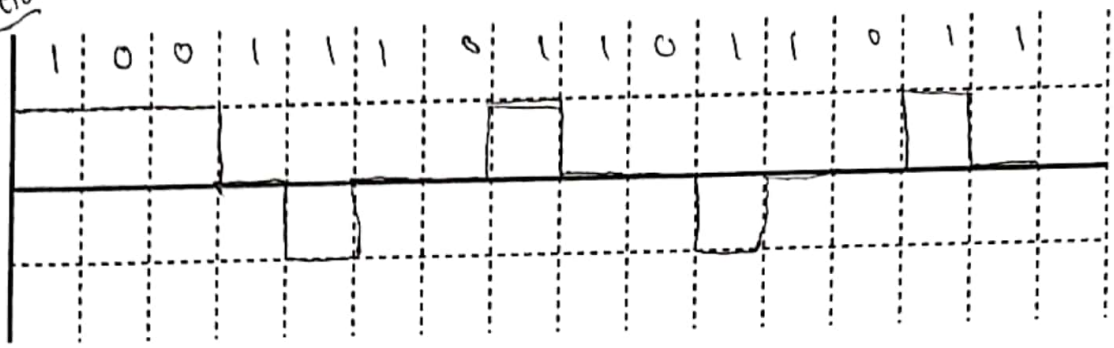
c)



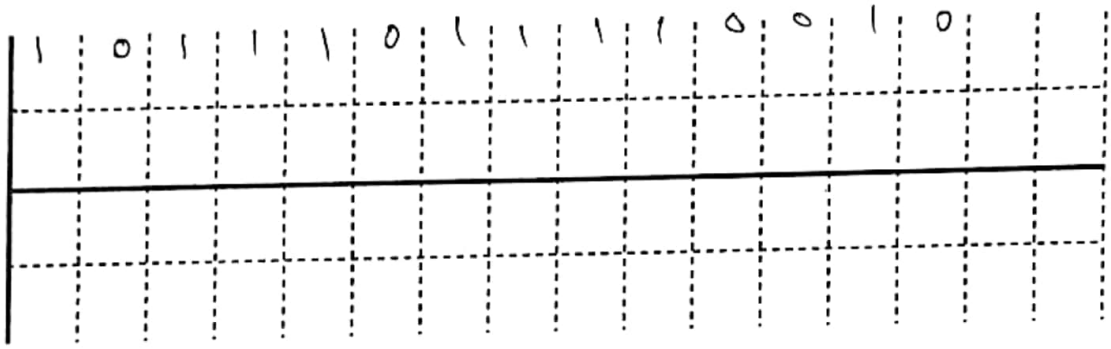
NO question

a)



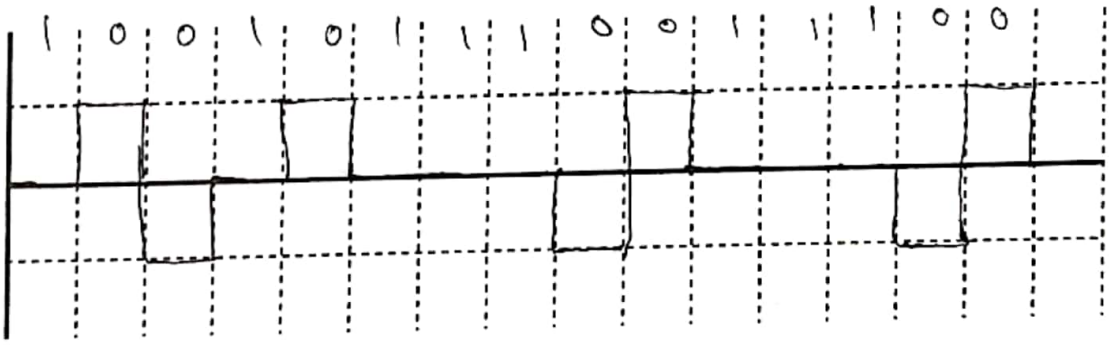
D-MAN

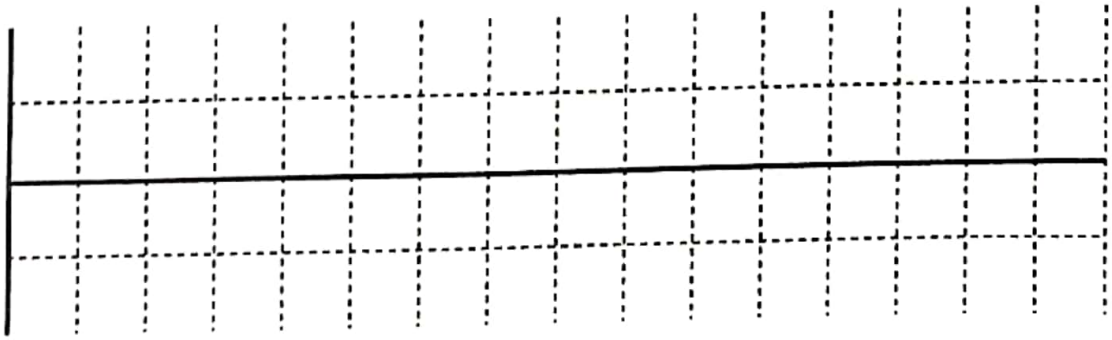
b).



pseudo temary

c)





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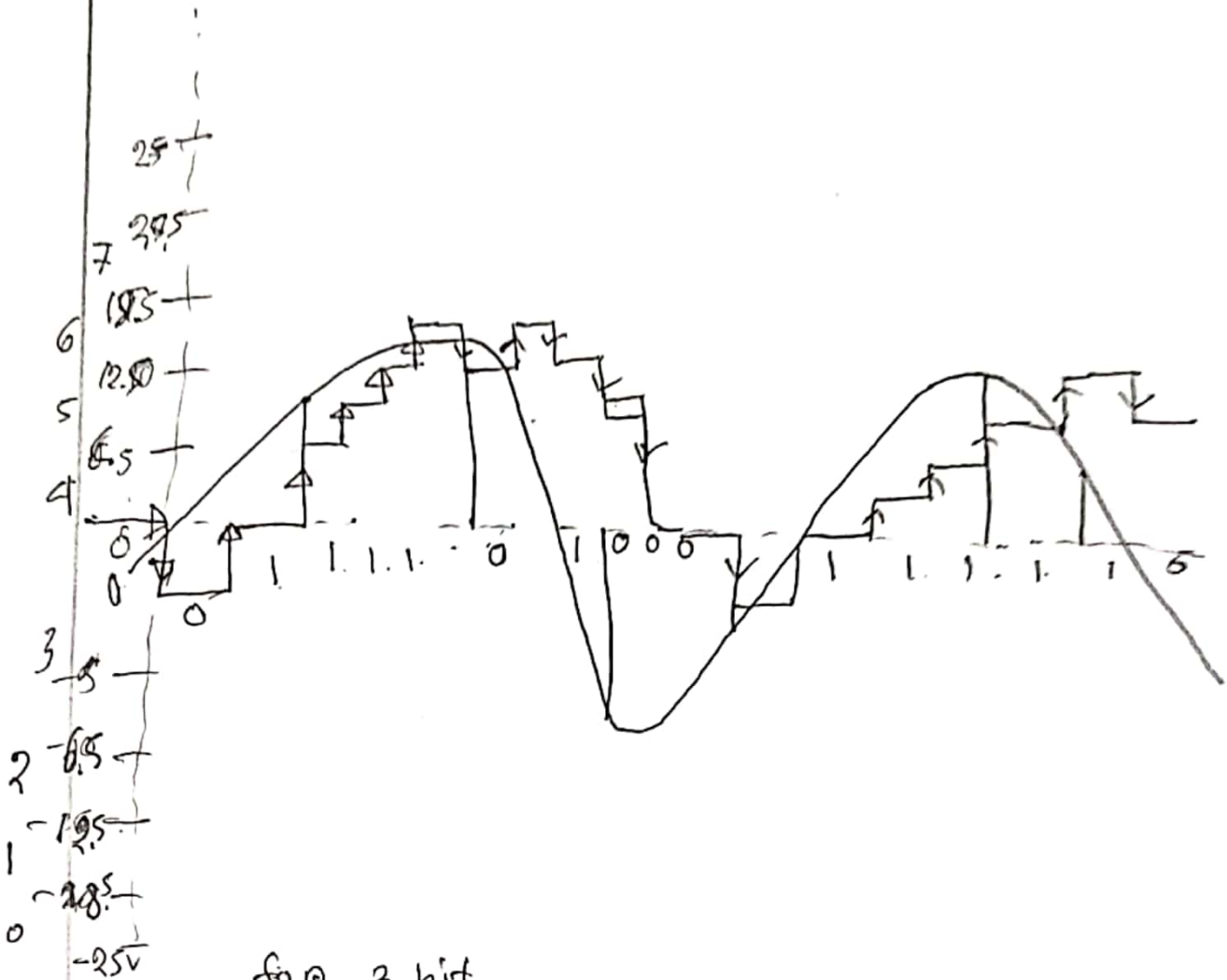
Answer to the Question-10



for 3 bit

2^K

$= 2^3$

$= 8$ levels

$$\Delta = \frac{25-(-25)}{8}$$

$$= \frac{50}{8}$$

$$= 6.25V$$

Normalize quantized value = ~~-20V (-2V)~~ -15V, 21.875 V,

= ~~5V~~ -9.375V, -15V, 21.875 V.